Kai Halpin

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Education

University of California, Irvine Expected June 2028

Bachelor of Science, Mechanical Engineering | GPA: 3.75

Awards/Honors: Recipient of over $100,000 in nationally competitive STEM scholarships: AIST Steel Intern Scholarship, Rose Hills Foundation Scholarship, GMiS Scholar, ERC Eco Scholarship, ACEC Scholarship, Roy Post Foundation Scholarship

Experience

Mechanical Engineering Intern, Pacific Steel Group – Mojave, CA June 2026 – Aug 2026

- Analyzed P&IDs for RO water treatment facility and roll shop, identifying 30+ key discrepancies between coordination model and P&IDs involving instrumentation, lockout/tagout procedures, equipment sizing, and automation systems.
- Performed field walkdowns to verify installation progress, validate design documentation, and identify discrepancies between engineering drawings and field conditions.
- Reviewed vendor drawings, equipment datasheets, and technical specifications to support punch list development.
- Performed structural load calculations for reverse osmosis water treatment facility pipe racks and rail systems to verify adequate equipment support and design requirements.

Energy Efficiency Engineer, UCI SMART Industrial Assessment Center – Irvine, CA Dec 2025 – Present

- Conduct DOE sponsored industrial energy assessments for small and medium-sized manufacturers to identify opportunities for improved energy efficiency and productivity.
- Execute assessments of plant operations, identifying over $25,000 of savings opportunities/implementation costs for energy efficiency measures for building retrofit projects including lighting, HVAC, and renewable energy systems.
- Deliver recommendations with an average two-year payback period, enabling manufacturers to reduce operating costs while advancing national sustainability goals.

Mechanical Engineering Intern, Sentinel Peak Resources – Inglewood, CA June 2024 – Aug 2024

- Analyzed Injection Profiles for over 250 wells over a 5-year historical span, identifying key Mechanical and Loss Intervals, contributing to data driven insights for improved system performance and reliability.
- Designed and implemented a fully automated data management system using Excel, integrating well survey diagrams into a centralized database, optimizing data flow and enhancing accessibility for engineering teams.
- Conducted analysis using technical data, finding actionable maintenance solutions to improve mechanical integrity.
- Identified and documented approximately 1,200 Problem Injection Intervals across 193 wells, including 103 injection levels potentially requiring repairs, providing insights for strategic field interventions.

Finance Director, Bayar Fellowship – Remote June 2024 – Mar 2026

- Directed team of 10+ in managing the financial operations of a 501c3 nonprofit focused on providing leadership and scholarship access to underrepresented high school students.
- Spearheaded a strategic merger with another nonprofit, expanding the organization's projected reach to 250,000+ students and increasing access to financial literacy and educational resources.
- Raised $8,000+ in scholarship funding through grant applications, donor outreach, and corporate partnerships.
- Collaborated with partner organizations to develop internship programs, creating career development opportunities.

Projects

PNEUMATIC POWERED RACK & PINION ROBOT Engineering Portfolio Link

- Designed electrical schematics and soldered, wired, and integrated all electrical & pneumatic subsystems.
- Implemented a PID controller with magnetometer feedback for autonomous navigation and heading correction.

Skills

Software: SolidWorks, Fusion 360, NavisWorks, MATLAB, Python, Power Automate Desktop, Excel, Bluebeam Revu.

Hardware: Precision Measurement Tools (Calipers, Micrometers), Machine Shop Equipment (Mill, Lathe, Drill Press, Band Saw), Tapping & Threading, 3D Printing, Lockout/Tagout (LOTO), Soldering, Electrical Prototyping, Mechanical Assembly.